

Diamond Price Analysis

~Exploratory Data Analysis (EDA) Summary Report

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Objective

The objective of this project was to analyze the factors affecting diamond prices using Exploratory Data Analysis (EDA). The project focused on cleaning the dataset, handling missing values and duplicate records, detecting outliers, analyzing categorical and numerical features, calculating statistical measures, and visualizing relationships between different diamond characteristics and their prices.

Dataset Information

Dataset Name: Diamonds Dataset

Source: Kaggle

Dataset Link:

<https://www.kaggle.com/datasets/shivam2503/diamonds>

Dataset Description

The Diamonds dataset contains information about different diamond characteristics, including carat weight, cut, color, clarity, depth, table percentage, dimensions, and total sales price. The dataset is useful for understanding how various physical and quality-related features influence the price of diamonds.

Data Exploration

The dataset was explored using functions such as `head()`, `shape`, `info()`, and `describe()`. These functions helped understand the dataset structure, identify data types, examine the number of records, and generate descriptive statistics before performing further analysis.

Data Cleaning

The dataset was checked for missing values and duplicate records. Missing values were handled wherever required, duplicate records were removed, and the dataset was verified for consistency. These preprocessing steps improved the quality of the data and prepared it for further analysis.

Statistical Analysis

Descriptive statistical measures such as mean, median, mode, variance, and standard deviation were calculated for the numerical features. These measures provided insights into the distribution, central tendency, and variability of different diamond characteristics.

Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method. Box plots were used to identify extreme values in numerical columns such as carat weight, depth percentage, table percentage, and total sales price. Identifying outliers helped understand unusual observations that could influence the analysis.

Feature Analysis

Both categorical and numerical features were analyzed to understand their impact on diamond prices. Categorical features such as cut, color, and clarity were compared, while numerical features such as carat weight and dimensions were studied to identify their relationship with the total sales price.

Correlation Analysis

Correlation analysis was performed to study the relationship between numerical variables. A correlation heatmap was created to identify strongly related features. The analysis showed that carat weight has a strong positive relationship with the total sales price, making it one of the most influential factors affecting diamond pricing.

Data Visualization

Several visualizations were created to better understand the dataset. These included Bar Chart, Line Chart, Histogram, Scatter Plot, Correlation Heatmap, and Box Plot. The visualizations helped identify distributions, feature relationships, and pricing trends among different diamonds.

Key Insights

The dataset was successfully cleaned before analysis by handling missing values and duplicate records. Statistical analysis provided a better understanding of the distribution of diamond characteristics. Outliers were identified using the IQR method, and correlation analysis revealed that carat weight has a significant influence on the total sales price. Visualizations clearly illustrated how different diamond features affect pricing.

Conclusion

This project provided practical experience in performing Exploratory Data Analysis using the Diamonds dataset. It covered data cleaning, statistical analysis, feature analysis, outlier detection, correlation analysis, and visualization techniques. The project demonstrated how EDA can be used to understand pricing patterns and identify the factors that influence diamond prices.

Tools Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Colab
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Project Outcome

Through this project, I gained hands-on experience in:

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Statistical Analysis
- Feature Analysis
- Correlation Analysis

- Outlier Detection
 - Data Visualization
 - Interpreting Analytical Results
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